

Syn-CXR RISE Score

Evaluator's Guide for Synthetic Chest Radiograph Assessment

Overview

The Syn-CXR RISE Score is a structured, five-point binary evaluation system designed to assess the radiological accuracy of synthetic chest radiographs generated from structured text prompts. Each of the five domains is scored independently as either 1 or 0, yielding a composite score ranging from 0 to 5.

Scoring Domains at a Glance

Domain	Criterion		Score 1	Score 0
1	Anatomical Validity	Does the generated background image demonstrate anatomically plausible structures?	Only anatomically plausible thoracic structures without distortion.	One or more implausible, distorted, or missing major anatomical structures.
2	Pathology Presence	Does the generated image demonstrate any abnormal findings suggestive of the pathology (finding)?	The prompted pathology (finding) is clearly visible.	Pathology (finding) is absent or different from the prompt.
3	Internal Consistency	Is the observed generated pathology (finding) radiologically plausible and realistic (natural)?	Imaging findings are consistent with the prompted pathology (finding) and follow natural radiological patterns.	Imaging findings show unrealistic features that do not naturally occur.
4	Attribute Concordance	Is the attribute/ location and side of the pathology (finding) depicted in the generated image correctly?	Attribute/ location and side are correctly depicted.	Wrong attribute, location, or side.
5	Similarity Index	Does the generated background image look the same as the reference image, regardless of the generated pathology (finding)?	The background image looks the same as the reference image.	The background image looks different from the reference image.

Understanding the Prompt Structure

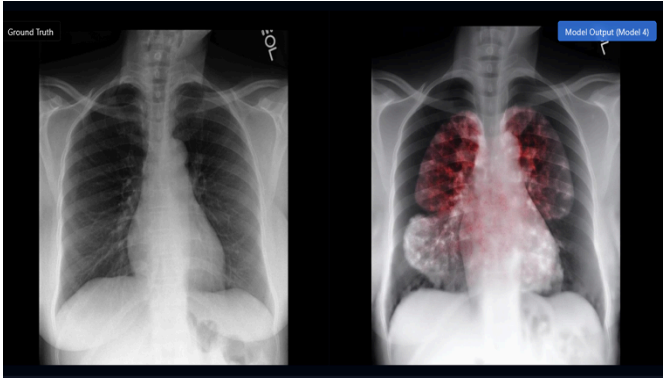
Each evaluation prompt comprises two components: an Attribute (specifying side, degree, or location) and a Finding (the radiological abnormality). Evaluators must parse both components before scoring, as distinct domains assess each component independently.

Full Prompt	Attribute	Finding
Right 6th rib fracture	<i>Right 6th rib</i>	<i>Fracture</i>
Mild cardiomegaly	<i>Mild</i>	<i>Cardiomegaly</i>
Right moderate pleural effusion	<i>Right moderate</i>	<i>Pleural effusion</i>

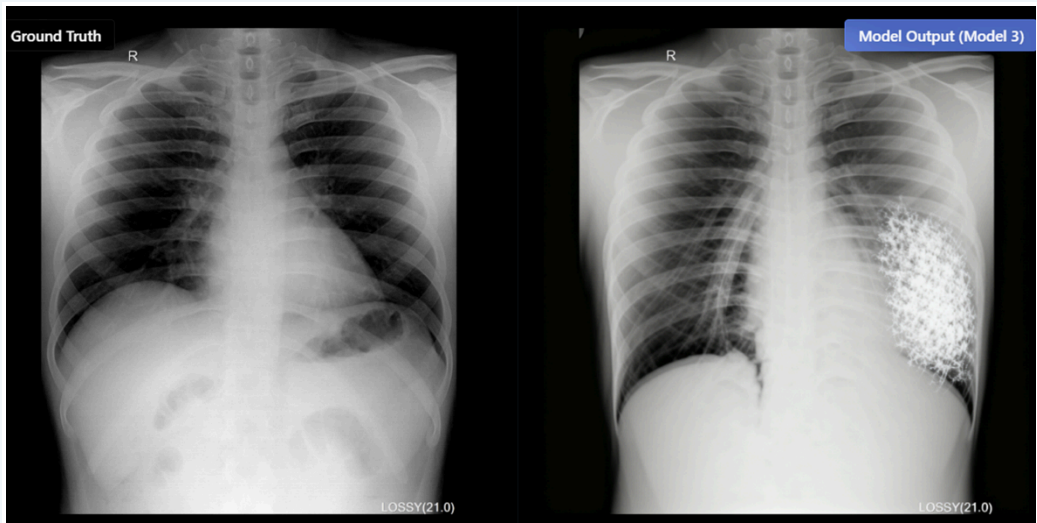
Step-by-Step Evaluation Workflow

Apply the five domains sequentially in the order presented below. Each domain has a defined scope.

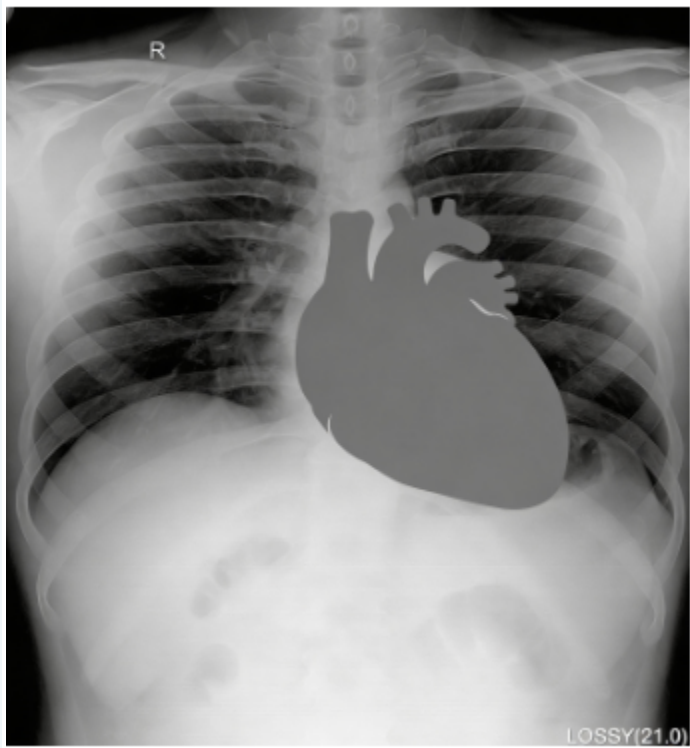
Domain-by-Domain Reference

1. Anatomical Validity	
Evaluate only the generated image. The reference image and any pathology region must be ignored entirely.	
Score 1 Only anatomically plausible thoracic structures without distortion.	Score 0 One or more implausible, distorted, or missing major anatomical structures.
Evaluate: - Lung fields, ribs, clavicles, and scapulae - Cardiomediastinal silhouette — shape and position - Diaphragm contours and bowel shadows	Do not assess: - The reference image - The pathology region
Illustrative Examples Example 1: Broncho-vascular markings are absent → Anatomical Validity = 0	
	
Example 2:	

Hilar structures and right hemidiaphragm are distorted → Anatomical Validity = 0



Example 3: Prompt — Mild cardiomegaly
Anatomical Validity = 1; thoracic anatomy (excluding the pathology region) is normal.
Pathology Presence is 1 and internal consistency is 0.



2. Pathology Presence	
Evaluate only whether a pathology (finding) is present in the generated image. DO NOT assess whether the pathology (finding) is radiologically correct.	
Score 1	Score 0

The prompted pathology (finding) is clearly visible.	Pathology (finding) is absent or different from the prompt.
Evaluate: • Presence of abnormality	Do not assess: • Whether it is radiologically correct • Whether secondary signs are present • Whether the diagnosis is accurate

3. Internal Consistency

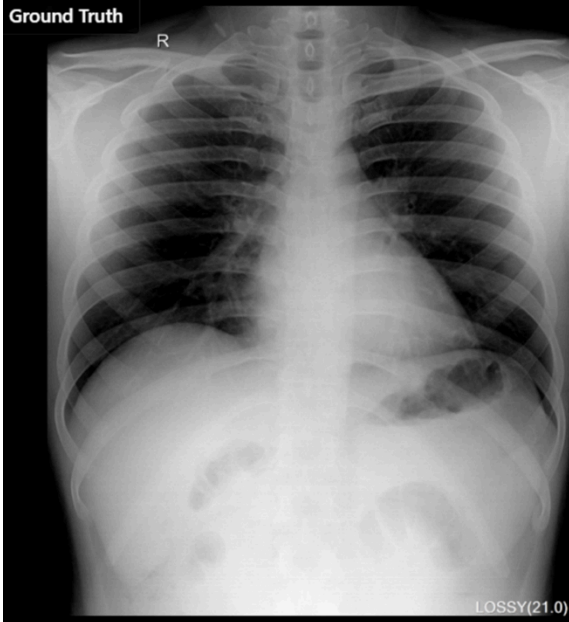
Evaluate whether the pathology (finding) is radiologically correct

Score 1 Imaging findings are consistent with the prompted pathology (finding) and follow natural radiological patterns.	Score 0 Imaging findings show unrealistic features that do not naturally occur.
Evaluate: • Radiological pattern consistent with the prompted finding • Presence or absence of expected secondary signs	

Illustrative Examples

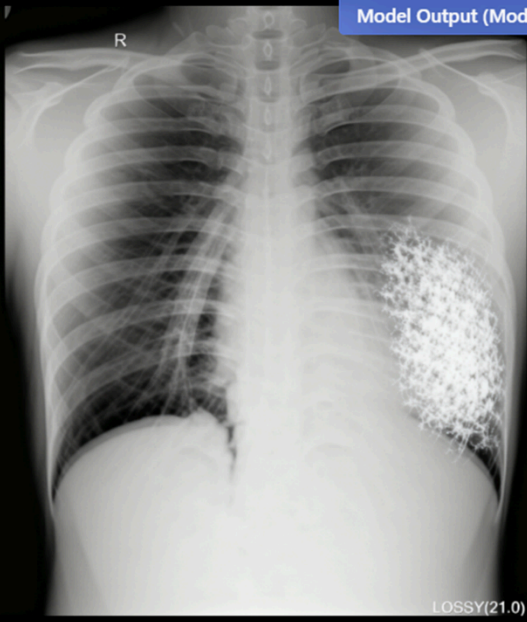
Example 1: *Prompt — Left lower zone consolidation*
Pathology Presence = 1; Internal Consistency = 0

Ground Truth



LOSSY(21.0)

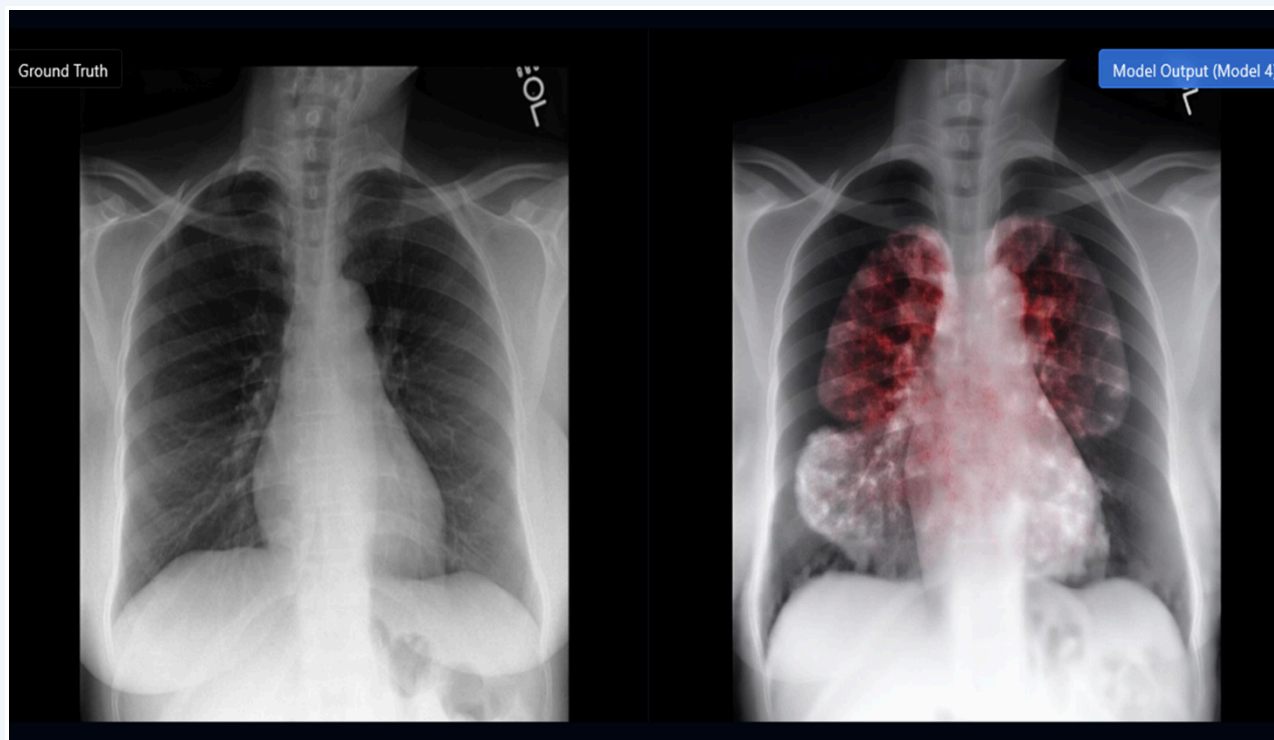
Model Output (Model 3)



LOSSY(21.0)

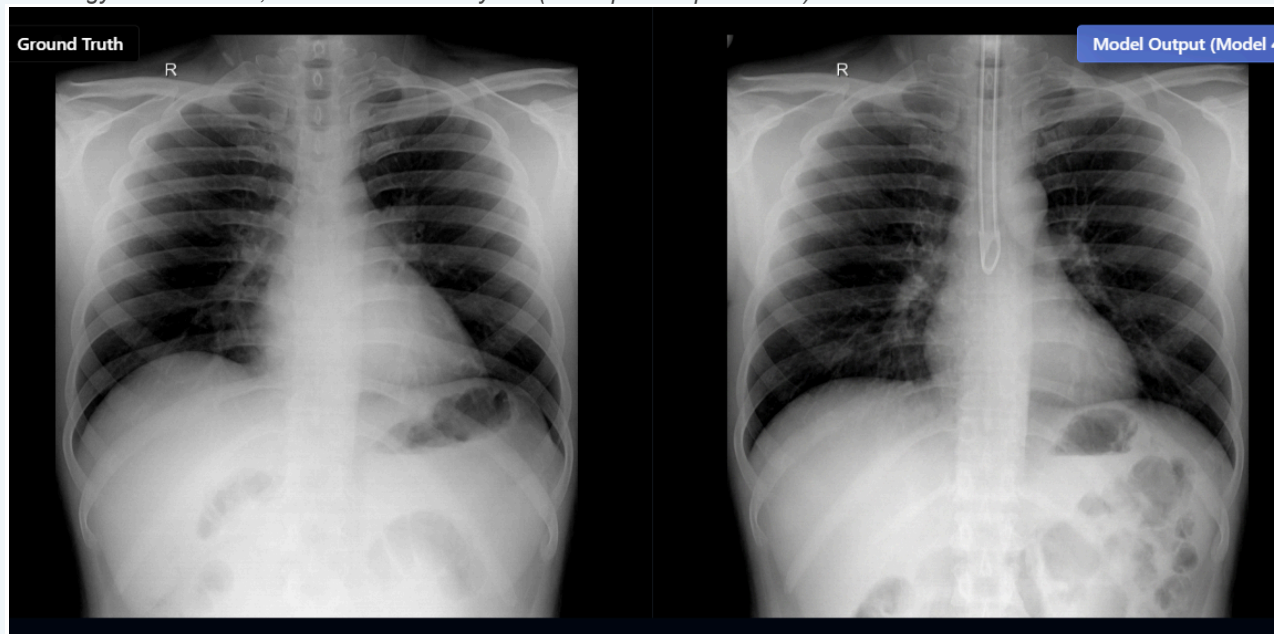
Example 2: *Prompt — Pulmonary oedema*

Pathology Presence = 0; Internal Consistency = 0



Example 3: Prompt — Normally positioned endotracheal tube

Pathology Presence = 1; Internal Consistency = 0 (tube tip is malpositioned)



4. Attribute Concordance

Evaluate whether attribute/ location and side of the prompted pathology (finding) are accurate.

Score 1 Attribute/ location and side is correctly depicted.	Score 0 Wrong attribute, location, or side.
Evaluate: • Laterality (right vs. left) • Location or zone (e.g., lower lobe, 6th rib) • Qualitative attribute (e.g., mild, moderate, large)	

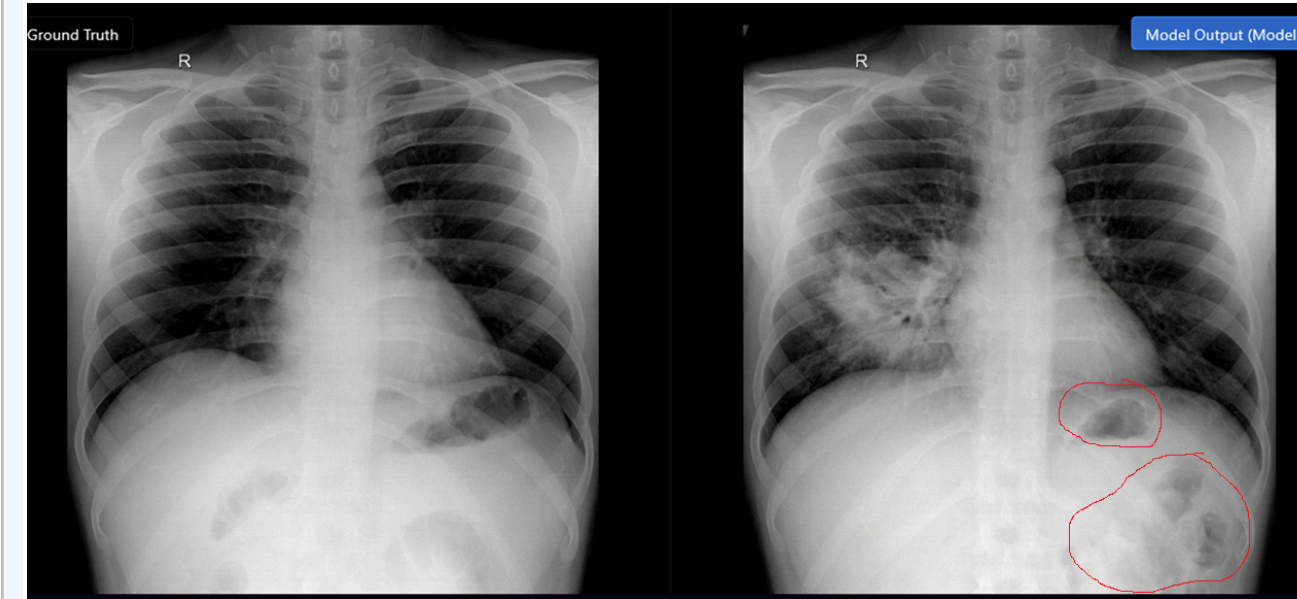
5. Similarity Index

Evaluate only the background features (*Ignoring the generated pathology*) of the generated image **COMPARED** to the reference image.

Score 1 The background image looks the same as the reference image.	Score 0 The background image looks different from the reference image.
Evaluate: • Same Radiographic marker • Same Breast shadow • Same Bowel/stomach gas • Presence of additional hallucinated artefacts • Increased or decreased graininess • Anything that helps you know that the background chest xray does not belong to the same patient.	Do not assess: • The pathology region

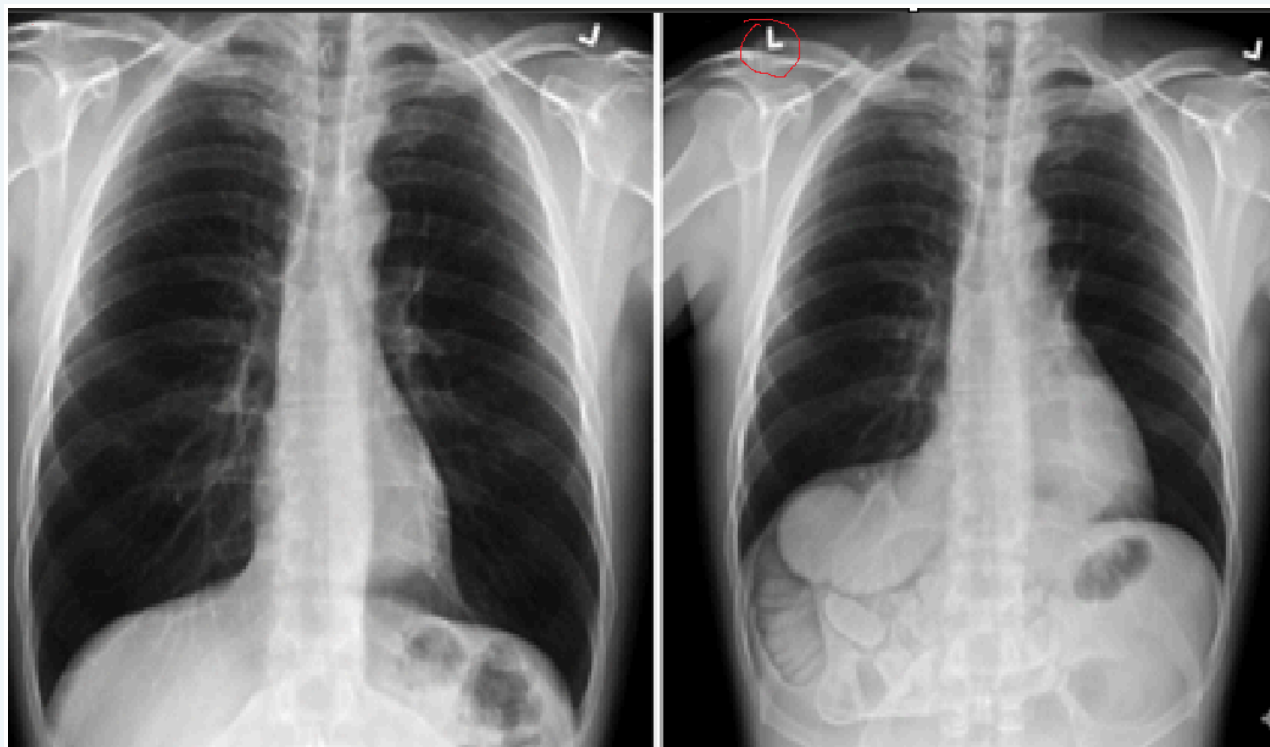
Illustrative Examples

Example 1:
Bowel shadows are altered → Similarity Index = 0



Example 2:

Radiographic markers and bowel gas pattern differ from reference → Similarity Index = 0



Example 3:

Hallucinated artefacts are present in the background → Similarity Index = 0

